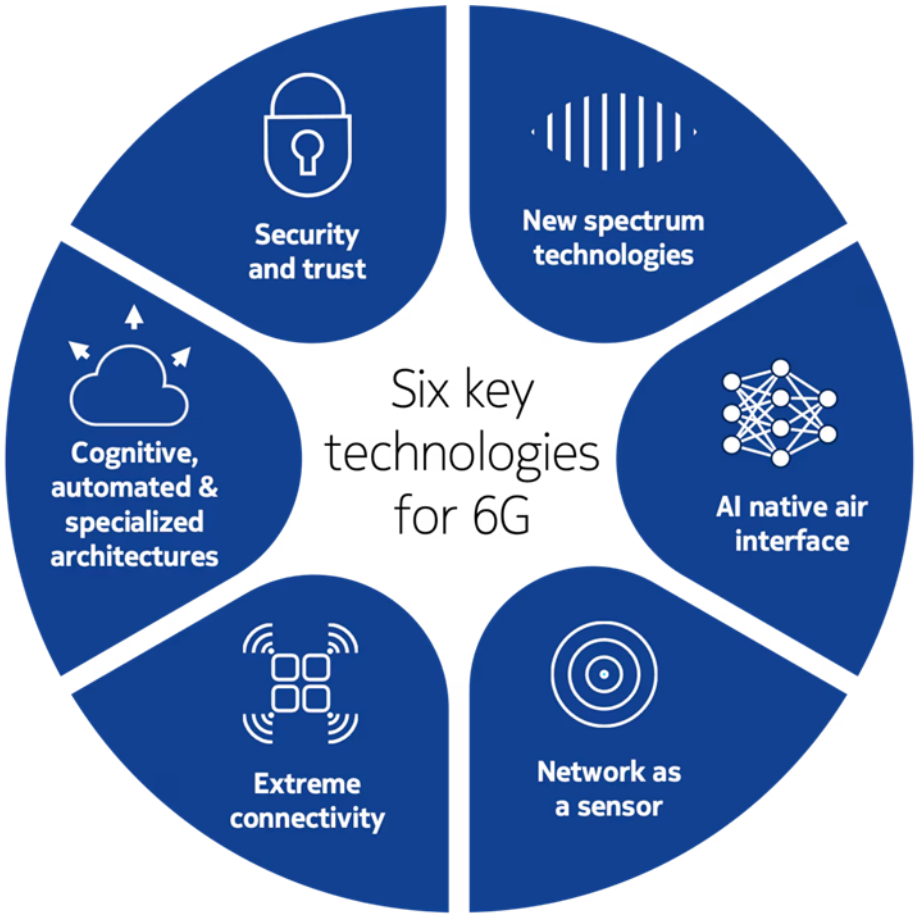
Optimizing Edge-AI for 6G via Split and Federated Learning Paradigms

MECT Thesis

- *Bruno Gomes nº103320*
- *Supervisor: Arnaldo Oliveira*
- *Co-supervisor: Fábio Coutinho*

universidade de aveiro

Context



Source: [1]

6G networks need to support

Holographic comms.	Low-latency Industrial IoT	Ubiquitous connectivity	Autonomous vehicles

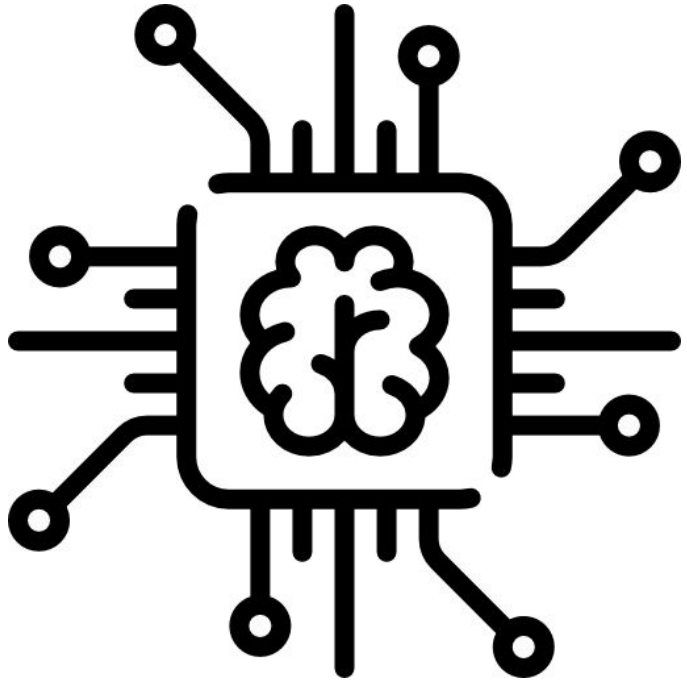
This means 6G networks must **cut latency**, **boost bandwidth** and **protect data** better than 5G

AI will be **crucial**: networks must be **dynamic** & **intelligent**

Problem

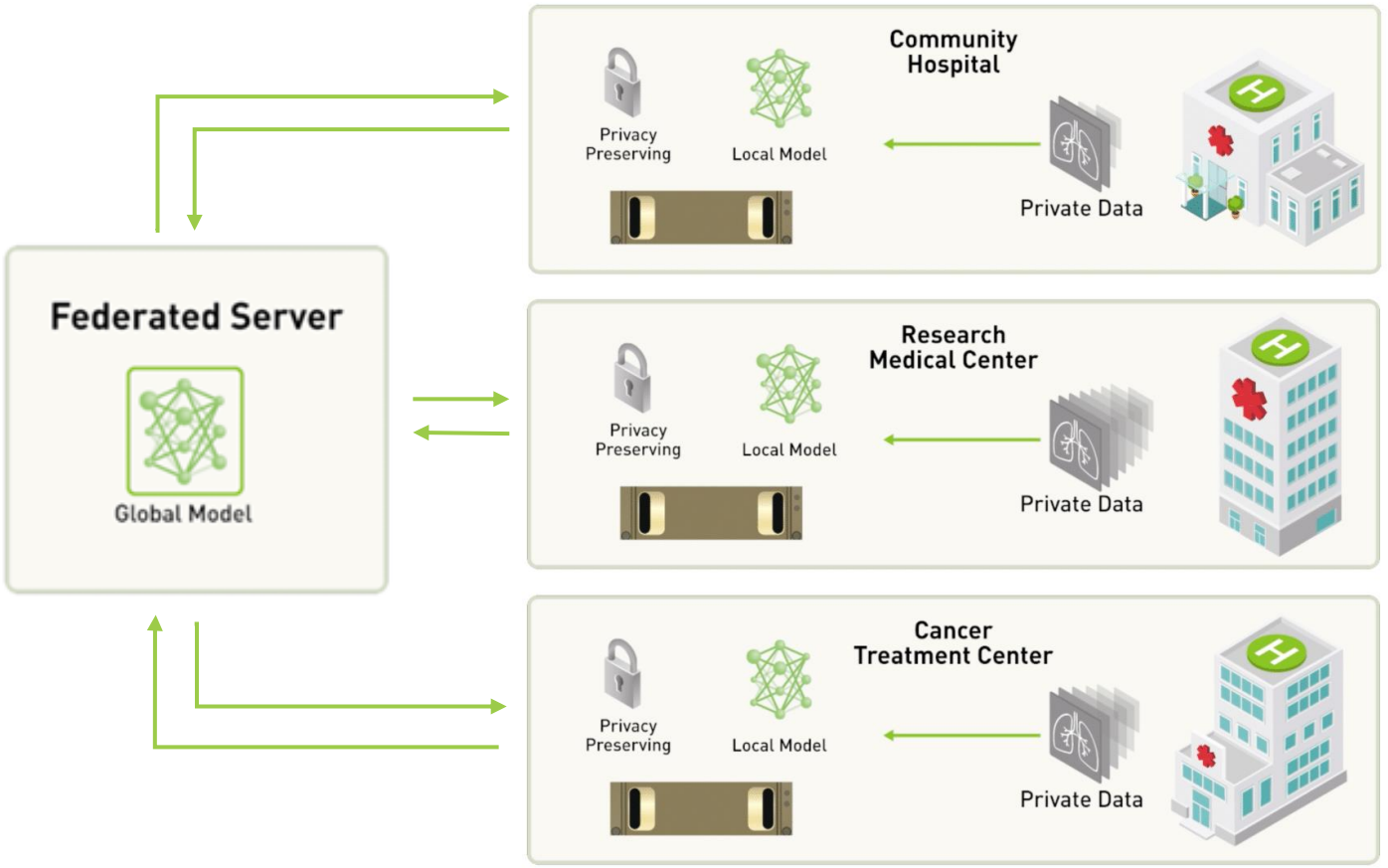
Traditional **centralized AI won’t scale** with 6G **data & devices**

This is where “**connected intelligence**” steps in



Solutions for 6G Networks Applications

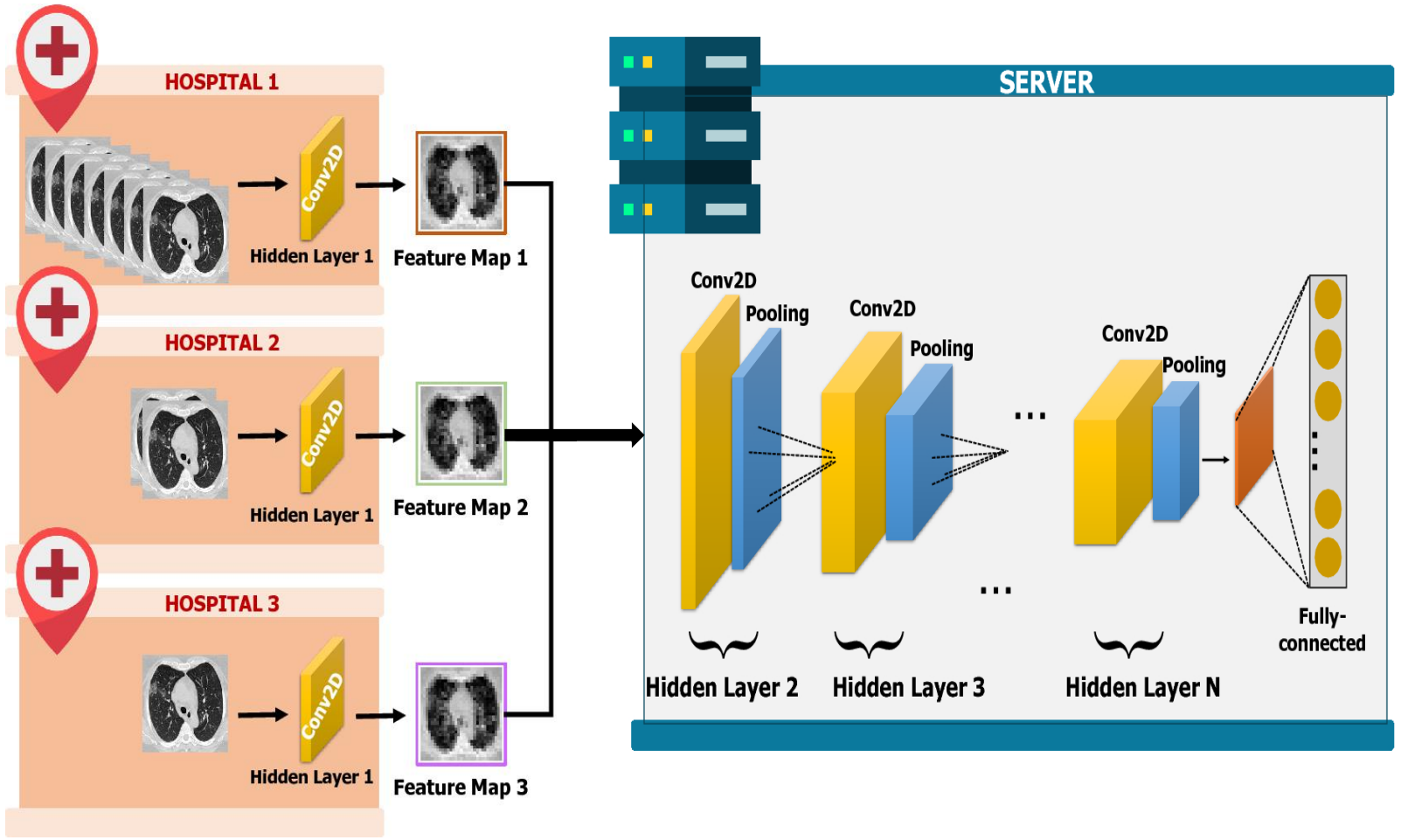
- **Federated Learning**
  - Devices train **locally** on their own data
  - Send only **model updates** (not raw data) to central server
  - Server **aggregates** updates into a global model



A centralized-server approach to federated learning [2]

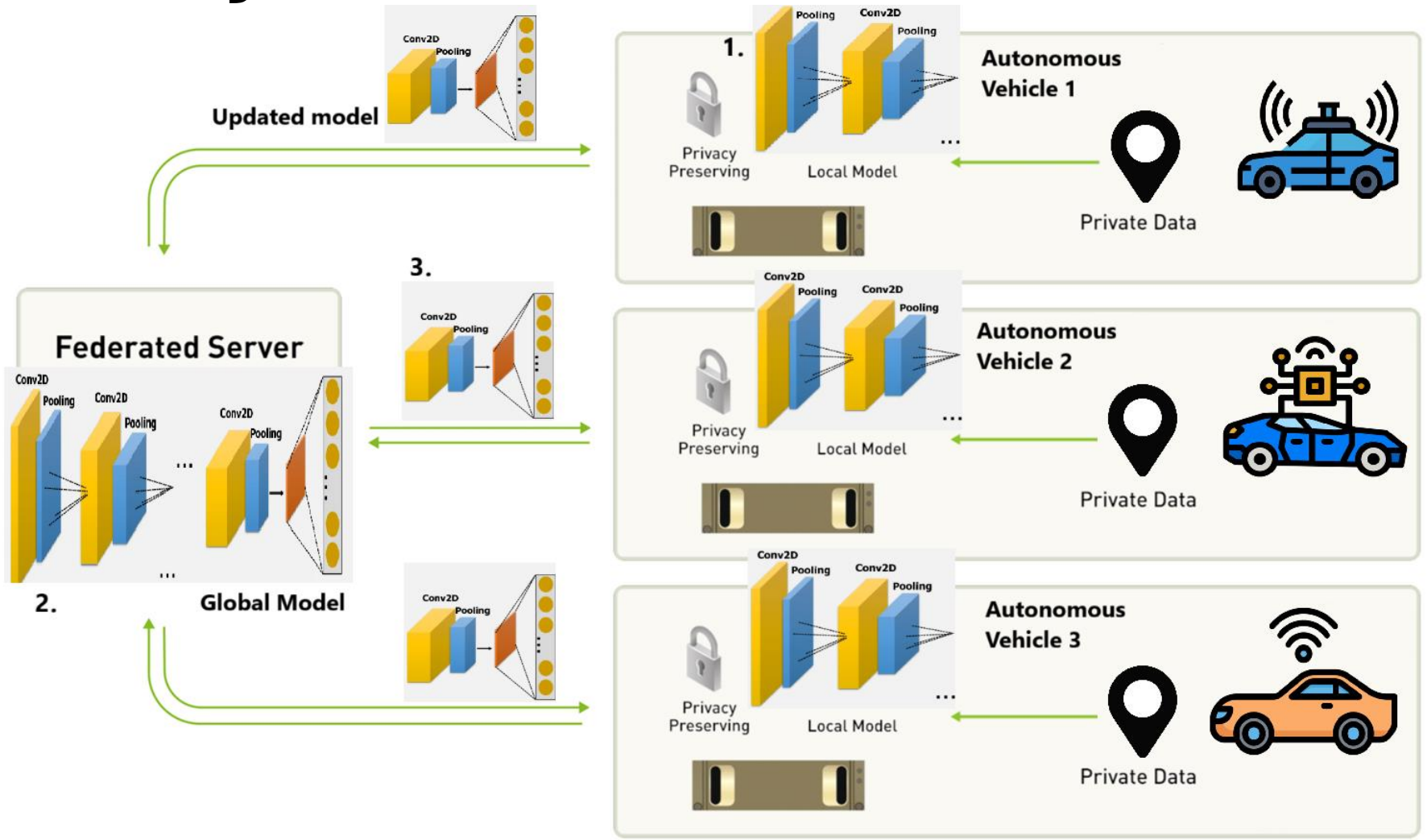
Solutions for 6G Networks Applications

- **Split Learning**
  - Devices run the **first layers** of the model
  - Send only **intermediate features** to server
  - Server runs the **remaining layers** of the model



Example of a framework for a multi-site split learning algorithm [3]

Use case: SFL-enabled vehicle mobility prediction system



Conclusion



Source: [4]

Thesis goal

- Implement and **test federated and split learning** on several edge devices to simulate **heterogenous 6G networks**

Key Outcomes

- Demonstrate how **AI improves network performance**
- Show improvements in **latency, bandwith and privacy**
- Provide **pratical use cases** where edge intelligence adds value

References

[1] Nokia, “Six key technologies for 6G,” [Online]. Available: https://www.nokia.com/6g/6g-explained/. [Accessed: Sep. 30, 2025].

[2] Nvidia, “Centralized-server approach to federated learning,” [Online]. Available: https://blogs.nvidia.com/blog/what-is-federated-learning/. [Accessed: Sep. 30, 2025].

[3] Y. J. Ha, G. Lee, M. Yoo, et al., “Feasibility study of multi-site split learning for privacy-preserving medical systems under data imbalance constraints in COVID-19, X-ray, and cholesterol dataset,” Sci. Rep., vol. 12, p. 1534, 2022, doi: 10.1038/s41598-022-05615-y.

[4] SpectrumIEEE, “6G Global Networks,” [Online]. Available: https://spectrum.ieee.org/examining-the-impact-of-6g-telecommunications-on-society. [Accessed: Sep. 30, 2025].


Questions?